

The EnvisionBOX Gesture Classification Challenge: A Multi-Corpus Benchmark for Co-Speech Gesture Detection via Privacy-Preserving Kinematic Extraction

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Co-speech gesture research is fragmented: laboratories produce richly annotated corpora in isolation, models rarely generalise beyond the corpus they were trained on, and cross-study comparison is structurally impossible without a shared benchmark. The primary barrier to pooling data across institutions is privacy: raw video of human participants cannot be freely shared under GDPR and equivalent regulations. We address both problems together. We present **EnvisionBOX**, the first multi-corpus, multi-language benchmark for co-speech gesture classification, built entirely on anonymised kinematic data extracted by **MaskBench**, an open-source benchmarking framework that processes video into skeletal landmarks—without releasing raw frames, audio, or face mesh. EnvisionBOX aggregates 32,575 annotated gesture clips from seven corpora, four languages, and five interaction settings (narration, conversation, direction-giving, clinical interview, and public presentation), covering 150+ unique speakers. We describe a speaker-independent evaluation protocol and two competitive baselines: a LightGBM model achieves 75.5% cross-validated and 69.5% held-out balanced accuracy; a 1D residual CNN achieves 66.5% and 60.1% respectively. The persistent gap between cross-validated and held-out accuracy demonstrates that cross-corpus gesture detection is a genuinely open challenge. We release training data, an evaluation script, and a starter kit, and invite the MINT community to build the shared benchmark infrastructure that gesture research has lacked.

1 Introduction

Co-speech gesture is a fundamental modality of face-to-face communication (McNeill, 1992; Kendon, 2004). Speakers gesture spontaneously and universally, and their gestures carry semantic, pragmatic, and interpersonal information that complements the spoken signal. Understanding,

modelling, and generating gesture in naturalistic interaction is therefore a central problem for multimodal NLP—and a core concern of the MINT research community.

Despite this importance, gesture research lacks the comparative infrastructure that has driven progress in neighbouring fields. In natural language processing, shared tasks—CoNLL named entity recognition (Tjong Kim Sang and De Meulder, 2003), SemEval semantic similarity, WMT machine translation—created the conditions for rapid community progress: common training and test splits, agreed evaluation metrics, and a public record of which systems work and why. Gesture research has none of this. Laboratories collect and annotate corpora independently, using different gesture taxonomies, different recording conditions, and different segmentation criteria. Models trained on one corpus generalise poorly to another, results cannot be directly compared across studies, and the field re-solves the same annotation and evaluation problems repeatedly.

The root cause is privacy. Video recordings of human participants carry identifiable facial, vocal, and behavioural information. GDPR and equivalent regulations in Germany, the Netherlands, China, and elsewhere prevent laboratories from sharing raw recordings across institutional boundaries without extensive ethical review, consent re-negotiation, and data processing agreements. In practice, this means each lab works with its own data in isolation.

Two recent developments make it possible to break this deadlock. First, pose estimation has crossed a practical threshold. Frameworks such as MediaPipe (Lugaresi et al., 2019) extract full-body 3D landmark coordinates from video at real-time speed on commodity hardware. The resulting world landmark sequences—joint positions in metric space, centred on the hips—carry no identifiable facial or vocal information and can be shared without the privacy risks of the original record-

ings. Second, video de-identification and pose evaluation tooling has matured: MaskBench (Riedel et al., 2025) (<https://maskbench.github.io/maskbench/>) provides a systematic framework for benchmarking pose estimators under different masking strategies, producing anonymised skeleton animations and kinematic feature files as its primary output.

We exploit both developments to build **EnvisionBOX**, the first cross-corpus shared benchmark for co-speech gesture detection. Our contributions are threefold:

1. **MaskBench as a privacy-preserving data donation pipeline.** We show that the MaskBench framework, run on raw video, produces anonymised world landmarks that preserve sufficient kinematic information for gesture detection while eliminating identifiable content. Gaussian blurring retains an average of 86% of keypoint accuracy across pose estimators compared to raw video, establishing the pipeline’s suitability for cross-lab data release (§4).
2. **The EnvisionBOX dataset.** Seven corpora, four languages, 32,575 annotated clips, 150+ speakers across five interaction settings, released under CC BY-NC-SA 4.0 as anonymised kinematic files (§5).
3. **A challenge design and strong baselines.** A speaker-independent evaluation protocol, balanced accuracy as primary metric, and two baselines that set a competitive bar while exposing the cross-corpus generalisation gap (§6–7).

2 Related Work

Gesture corpora. The field has produced a number of richly annotated corpora, including the SaGA corpora for German direction-giving dialogue (others, 2013, 2022a), ECOLANG for English narration (others, 2020), MULTISIMO for multi-party English conversation (others, 2021), and ZHUBO for Mandarin narration (others, 2022b), among others. Each uses a different recording setup, annotation scheme, and gesture taxonomy, making cross-corpus evaluation structurally difficult. No prior work has unified multiple annotated corpora under a common evaluation framework with a shared held-out test set.

Automatic gesture detection. Earlier gesture detection systems relied on hand-crafted kinematic features from motion capture or wearable sensors (Nyatsanga et al., 2023). MediaPipe-based pipelines have lowered the cost of kinematic extraction to the point where pose-based gesture detection from ordinary video is practical at scale (Lugaresi et al., 2019). The EnvisionHGDetector (Pouw et al., 2026) is a Python package providing a hybrid CNN–LightGBM pipeline for co-speech gesture detection trained on five of the corpora used here; it represents the state of the art on this task and serves as our primary baseline. Prior evaluations have been conducted within single corpora or on corpora from the same lab; no study has evaluated gesture detection on held-out speakers from multiple independent corpora simultaneously.

Privacy-preserving behavioural data. Video de-identification for behavioural research has received growing attention as GDPR enforcement has tightened. MaskAnyone (Owoyele et al., 2024) provides a multi-stage pipeline combining YOLOv11 detection, SAM2 segmentation (Ravi et al., 2024), and MediaPipe or OpenPose pose estimation to produce de-identified video with overlaid skeleton animations. The MaskingOPS framework (Owoyele et al., 2025) situates these tools within an end-to-end tutorial for behavioural researchers. MaskBench (Riedel et al., 2025) extends this ecosystem with systematic evaluation of how masking strategies affect downstream pose quality—the question that directly motivates our data release pipeline.

NLP shared tasks as a model. The CoNLL shared task series demonstrated that standardised evaluation on common data drives disproportionate progress relative to resource investment (Tjong Kim Sang and De Meulder, 2003). SemEval has applied the same model to a broad range of semantic and pragmatic phenomena. Gesture research is at an earlier stage than NLP was in 2003, but the same principle applies: a common benchmark with a held-out test set is the most efficient way to establish what works and why. EnvisionBOX is the gesture research community’s first step in this direction.

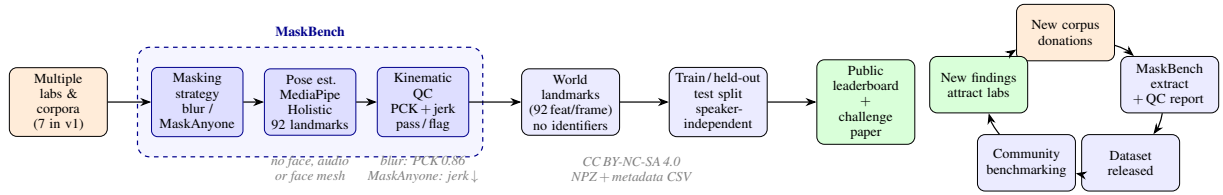


Figure 1: **(a) From lab data to challenge.** Raw annotated video from multiple laboratories enters the MaskBench pipeline (Riedel et al., 2025), which applies a masking strategy (Gaussian blur or MaskAnyone overlay), extracts MediaPipe world landmarks, and runs a kinematic quality gate (PCK and jerk) to pass or flag each corpus before release. Gaussian blurring retains PCK 0.86 on average; MaskAnyone variants achieve the lowest jerk, producing smoother tracks for gesture boundary detection. Only passing corpora are included; no raw video, audio, or face mesh is released. A speaker-independent train/test split is frozen; the held-out set is scored at challenge time against a public leaderboard. **(b) Virtuous cycle of data solicitation.** Challenge findings expose cross-corpus generalisation gaps and attract new corpus donations, each processed through the same MaskBench extraction and QC pipeline before release. The evaluation infrastructure is reused across rounds.

3 A Challenge-Based Approach to Gesture Benchmarking

Shared evaluation frameworks have been the primary engine of progress in NLP since at least the CoNLL-2003 shared task (Tjong Kim Sang and De Meulder, 2003). The mechanism is simple but powerful: a common held-out test set, an agreed evaluation metric, and a public record of which systems work and why. Gesture research is at an earlier stage, but the same logic applies. Figure 1(a) shows the ideal end-to-end workflow and Figure 1(b) shows how challenge rounds compound over time.

Ideal workflow. A community gesture benchmark proceeds in five stages. (1) **Corpus donation:** laboratories with gesture-annotated video register interest; MaskBench processes their recordings to world landmarks and verifies that no identifiable content remains. (2) **Curation:** challenge organisers harmonise labels, clip boundaries, and speaker-independent splits across donated corpora. (3) **Community engagement:** the training set is released openly; participants download data, train models, and run local evaluations using the published evaluation script. (4) **Evaluation:** the held-out test set is released at a fixed date; submitted predictions are scored centrally; a public leaderboard is published with per-corpus breakdowns. (5) **Knowledge integration:** the challenge paper documents what worked, what failed, and which interaction contexts remain hardest; findings are disseminated through ISGS, TDCC-SSH, and NLP venues, attracting new corpus donors for the next round (Figure 1(b)).

Challenges in this approach. Three structural difficulties must be addressed.

Label harmonisation. Gesture annotation is theory-laden. What counts as a GESTURE versus a MOVE differs across labs, annotation tools, and research traditions rooted in McNeill’s gesture typology (McNeill, 1992) or Kendon’s continuum (Kendon, 2004). Post-hoc harmonisation into three conservative categories (Section 5.2) is tractable but inevitably flattens annotation nuance; future rounds should explore finer-grained harmonised schemes with cross-lab calibration studies.

Corpus imbalance. ECOLANG contributes 54% of training clips in EnvisionBOX v1. Without countermeasures, models may implicitly learn features specific to English narration rather than general gesture kinematics. We address this with stratified splits; Phase 2 per-corpus leaderboards will make corpus-specific performance gaps explicit so future dataset versions can be better balanced.

Trust and sustainable donation. Corpus contribution requires confidence that data will be anonymised correctly, attributed fairly, and not misused. MaskBench’s verifiable anonymisation pipeline—no raw video, audio, or face mesh in released files—together with CC BY-NC-SA 4.0 licensing and the commitment to credit all contributors as co-authors addresses the trust barrier. Lowering the technical barrier further through automated MaskBench submission pipelines is a priority for future rounds.

4 MaskBench: Privacy-Preserving Kinematic Extraction

4.1 Framework Overview

MaskBench (Riedel et al., 2025) is the open-source toolchain that makes generating privacy-safe gesture challenges possible (<https://maskbench.github.io/maskbench/>). It is a framework that benchmarks pose estimators on raw and masked video. It wraps seven estimators—YoloPose (v11-l), MediaPipe Pose (heavy), OpenPose (BODY-25), and automated and human-in-the-loop variants of MaskAnyone—under a unified configuration interface. Given an input video, MaskBench runs inference with each estimator, computes both accuracy metrics (Percentage of Correct Keypoints, RMSE, Euclidean distance) against ground truth where available, and kinematic stability metrics (velocity, acceleration, jerk) to assess temporal smoothness. All output is stored in standardised JSON pose files.

For the EnvisionBOX dataset, we use the MediaPipe Holistic estimator run via MaskBench to extract **world landmarks**: 23 upper-body joints, each represented as $(x, y, z, \text{visibility})$, giving 92 features per frame in 3D metric space centred on the speaker’s hips. No raw video frames, audio, or face mesh coordinates are included in the released data.

4.2 Masking Strategies and Pose Quality

A key question for our data donation pipeline is whether privacy-preserving masking degrades the kinematic signal to the point where gesture detection is no longer possible. Table 1 reports average Percentage of Correct Keypoints (PCK, threshold 0.2) for five pose estimators applied to video masked with four strategies, relative to raw video predictions as reference.

Estimator	Blur	Pixel	Cont.	Fill
YoloPose	.95	.09	.93	.32
MediaPipe Pose	.95	.81	.56	.34
OpenPose	.88	.10	.62	.01
MaskAnyoneAPI-MP	.85	.30	.00	.00
MaskAnyoneUI-MP	.95	.63	.00	.07
Mean	.86	.23	.44	.11

Table 1: PCK (threshold 0.2) relative to raw video for four masking strategies: Gaussian blur, pixelation, contour overlay, and solid fill. Blurring retains 86% of keypoint accuracy on average; other strategies degrade performance substantially (Riedel et al., 2025).

Gaussian blurring retains the highest average PCK (0.86) because pose estimators rely more on learned body geometry and spatial priors than on fine-grained pixel texture (Riedel et al., 2025). Pixelation (0.23), contour overlay (0.44), and solid fill (0.11) degrade performance substantially and are not used in the EnvisionBOX pipeline. This result validates blurring as a masking strategy compatible with downstream kinematic analysis: the structural information needed for gesture detection—relative limb positions, wrist trajectories, arm velocity—is preserved in the extracted world landmarks.

4.3 From Video to Challenge Format

For each corpus video, the EnvisionBOX pipeline proceeds as follows: (1) MaskBench runs MediaPipe Holistic at model complexity 1, confidence threshold 0.5, with landmark smoothing enabled; (2) world landmark sequences are exported per clip as NumPy arrays of shape $(T \times 92)$; (3) skeleton animations (stick figures rendered onto contour-masked silhouettes, no facial detail) are generated for qualitative inspection; and (4) metadata (corpus, speaker ID, language, setting, label) is written to a companion CSV. No audio, face mesh, or raw video frames leave the pipeline.

5 The EnvisionBOX Dataset

5.1 Corpus Overview

EnvisionBOX aggregates seven independently collected and annotated corpora. All corpus providers have approved data donation for the challenge; full author details and corpus citations are being verified with each data owner before publication (see §9). Table 2 gives an overview.

The corpora span five interaction settings that vary in register, interactional structure, and degree of speaker constraint. Narration (ECOLANG, ZHUBO) and public presentation (TEDM3D) involve largely monologic speech; direction-giving (SaGA, SaGA++) and group discussion (MULTISIMO) involve multi-party interaction; clinical interviews (GESRes) introduce an asymmetric interactional frame. This diversity is intentional: a model that detects gesture only in one setting will fail on others.

5.2 Label Harmonisation

Each corpus was originally annotated using its own gesture taxonomy. We harmonise these into three

Corpus	Language	Setting	Clips	Speakers	License
ECOLANG	English	Narration	17,625	[TODO: N]	CC BY-NC-SA 4.0
MULTISIMO	English	Group discussion	4,202	[TODO: N]	CC BY-NC-SA 4.0
SaGA++	German	Direction-giving	2,911	[TODO: N]	CC BY-NC-SA 4.0
ZHUBO	Chinese	Narration	2,414	[TODO: N]	CC BY-NC-SA 4.0
GESRes	German	Clinical interview	2,392	[TODO: N]	CC BY-NC-SA 4.0
TEDM3D	English	Public presentation	1,551	[TODO: N]	CC BY-NC-SA 4.0
SaGA	German	Direction-giving	1,480	[TODO: N]	CC BY-NC-SA 4.0
Total	3 languages	5 settings	32,575	150+	CC BY-NC-SA 4.0

Table 2: The seven corpora comprising the EnvisionBOX dataset, ordered by clip count. Speaker counts and full corpus citations will be added after contributor verification (§9). ECOLANG contains adults only; MULTISIMO is restricted to non-commercial use per its original licence.

labels aligned with the framework used by the EnvisionHGDetecter (Pouw et al., 2026):

GESTURE A co-speech gesture with clear form and a communicative function that is coordinated with speech. Includes representational, metaphoric, deictic, and beat gestures (McNeill, 1992).

MOVE Purposeful limb movement not classifiable as a discrete communicative gesture: self-adaptors, postural adjustments, and beats with no recoverable semantic content.

NOGESTURE Rest position or non-communicative movement.

The resulting label distribution across 32,575 clips is approximately balanced: GESTURE 48.8% (15,896 clips), NOGESTURE 48.7% (15,844 clips), MOVE 2.6% (835 clips). The near-equal GESTURE/NOGESTURE balance reflects stratified sampling that corrects for corpus-level imbalance.

5.3 Clip Format and Metadata

Each clip is stored as a NumPy .npy file of shape $(T \times 92)$, where T is the clip duration in frames. The 92 features correspond to 23 upper-body MediaPipe world landmarks, each with four values: x , y , z (in metres, hip-centred) and visibility. Hand landmarks (21 keypoints per hand, 168 additional features) are available separately as an optional extension.

A companion metadata CSV provides, for each clip: clip_id, corpus, speaker_id, language, setting, label, and n_frames. Speaker IDs are anonymised within each corpus; no names or demographic information beyond language and corpus are included.

5.4 Train/Test Split

The split is **speaker-independent**: no speaker appears in both training and test partitions. Within each partition, clips are stratified by corpus, language, and label to ensure representative coverage. The 70% training set (approximately 22,800 clips) is released publicly with labels; the 30% test set ($\approx 9,800$ clips) is held out until Phase 2 of the challenge.

6 Challenge Design

6.1 Task

The primary task is **binary gesture detection**: given a variable-length world landmark clip, classify it as GESTURE or NOGESTURE. MOVE clips are folded into NOGESTURE for the primary track, following the convention in Pouw et al. (2026). An optional secondary track for three-class classification (GESTURE / MOVE / NOGESTURE) is planned for Phase 2.

6.2 Evaluation

The primary metric is **balanced accuracy**—the unweighted mean of per-class recall:

$$\text{Balanced Acc} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (1)$$

Balanced accuracy is insensitive to the near-equal class distribution and is directly comparable across models trained on different corpus subsets. Secondary metrics reported for all systems are per-class F1, sensitivity, specificity, and a full confusion matrix. An evaluation script implementing all metrics is publicly available; participants are encouraged to run it on their own validation splits before Phase 2 submission.

6.3 Submission Format

Participants submit a CSV with one row per test clip containing `clip_id` and `predicted_label`; an optional `gesture_probability` column enables calibration analysis. Docker image submission is supported for full reproducibility. **[TODO: Finalise submission platform: CodaLab / EvalAI / custom.]**

6.4 Phases

Phase 1 (Q3 2026). Training data, skeleton animations, metadata, evaluation script, and starter kit released. Community builds and evaluates models against their own held-out training splits. Community survey collects input on downstream task requirements.

Phase 2 (Q4 2026). Held-out test set released; final model submissions accepted; public leaderboard published with balanced accuracy rankings and per-corpus breakdowns.

Publication. A challenge description paper crediting all corpus contributors as co-authors will be submitted to **[TODO: venue TBC after MINT outcome]**.

7 Baseline Results

We report baselines from the EnvisionHGDetector v3.0.4 training pipeline (Pouw et al., 2026) run on the EnvisionBOX dataset with a speaker-independent 70/30 train/test split and Bayesian hyperparameter search (50 trials, 3-fold cross-validation on training set, balanced accuracy as objective).

7.1 Models

LightGBM (Ke et al., 2017). A gradient-boosted tree model operating on 100 hand-engineered features computed over a 5-frame sliding window: current pose (18 features), frame-to-frame velocity (18), wrist speeds (2), finger-to-finger distances (18), acceleration, trajectory smoothness, wrist height, arm spread, arm extension, bilateral symmetry, and per-keypoint visibility scores (20 total). Binary classification: GESTURE vs. NOGESTURE (MOVE merged into NOGESTURE).

1D Residual CNN. A temporal convolutional network with three residual blocks (48/96/192 filters), operating on raw 25-frame windows of the 92-dimensional world landmark representation. Dual-head output: a sigmoid motion detection head and

a softmax gesture classification head support both binary and three-class inference.

7.2 Results

Model	CV	Test	OOS
LightGBM (100 features)	75.5	69.4	75.0
LightGBM (L-92 raw)	75.7	65.3	—
1D Res. CNN	66.5	60.1	81.3

Table 3: Balanced accuracy (%) for two baseline models. **CV**: cross-validated on training set (speaker-independent 3-fold). **Test**: held-out test set balanced accuracy. **OOS**: out-of-sample accuracy on continuous recordings using a sliding-window inference regime—not directly comparable to clip-level test figures (see §7.3). **[TODO: Update CNN numbers once 100-trial sweep completes.]**

7.3 Analysis

The LightGBM model trained on 100 engineered features achieves the best clip-level test performance (69.4% balanced accuracy), while the model trained on raw 92-dimensional world landmarks achieves slightly lower test accuracy (65.3%) despite marginally higher cross-validated performance (75.7% vs. 75.5%). This pattern suggests that engineered features—which explicitly encode velocity, acceleration, and bilateral symmetry—generalise better to unseen speakers than raw landmark coordinates, which may overfit to speaker-specific resting poses and scale.

The CNN achieves lower clip-level accuracy than either LightGBM variant (60.1% test) but substantially higher out-of-sample accuracy on continuous recordings (81.3% OOS), measured on the MULTISIMO, SaGA, and external corpora using sliding-window inference. This reversal reflects a methodological difference: OOS accuracy is class-weighted on continuous streams, while clip-level test accuracy is balanced accuracy on isolated clips. The CNN’s temporal convolutional structure appears better suited to detecting gesture onsets and offsets in continuous speech than to classifying pre-segmented clips.

Critically, both models show a substantial gap between cross-validated and held-out performance (6.1 pp for LightGBM, 6.4 pp for the CNN). This gap persists even with speaker-independent cross-validation, confirming that cross-corpus generalisation—rather than within-corpus speaker generalisation—is the dominant challenge. Models must generalise across different languages,

recording setups, interaction types, and annotation conventions simultaneously. **[TODO: Add per-corpus accuracy breakdown to Table 3 once pipeline run is complete.]**

8 Discussion

Building the EnvisionHGDetector baseline and running it on the pooled EnvisionBOX corpus surfaced four findings that bear on how the community should design and interpret gesture detection benchmarks. We foreground them here rather than bury them in analysis notes.

Insight 1: engineered features generalise across corpora; raw landmarks overfit to recording conditions. LightGBM on 100 hand-crafted features (wrist velocity, bilateral symmetry, arm spread, per-keypoint visibility) achieves 69.4% held-out balanced accuracy, compared to 65.3% for the same model trained on raw 92-dimensional world landmarks—despite the raw model scoring marginally higher under cross-validation (75.7% vs. 75.5%). The inversion is instructive: within-corpus speaker-independent CV does not expose the generalisation gap that appears when the test corpus is drawn from a different recording context, language, and annotation convention. Explicitly encoding kinematic invariants (velocity, acceleration, bilateral ratios) provides a prior that transfers; raw coordinates encode speaker-specific scale, resting pose, and recording geometry that do not. Benchmark participants should report both CV and held-out figures; CV alone is misleading for cross-corpus tasks.

Insight 2: clip-level and stream-level accuracy rank models differently—these metrics are not commensurable. The 1D residual CNN achieves 60.1% balanced accuracy on held-out clips but 81.3% out-of-sample accuracy on continuous recordings under sliding-window inference. The LightGBM model reverses: stronger on clips, weaker on streams. The gap is not noise; it reflects a structural difference between the two tasks. Clip-level classification is a boundary-given problem: the model sees a pre-segmented window and assigns a label. Stream-level detection is a boundary-finding problem: the model must locate gesture onsets and offsets in continuous speech. The CNN’s temporal convolutional structure is better suited to the latter; the LightGBM’s feature engineering is better suited to the former. EnvisionBOX Phase 2

will add a streaming evaluation track specifically to surface this distinction and prevent participants from optimising the wrong objective for their intended use case.

Insight 3: cross-corpus generalisation is the binding constraint, not cross-speaker generalisation. The 6-percentage-point gap between cross-validated and held-out accuracy persists even with a fully speaker-independent CV protocol. This means the dominant source of generalisation failure is not speaker identity but the simultaneous shift in recording setup, language, interaction type, gesture taxonomy, and annotation convention that occurs when the test set is drawn from a different corpus. Standard speaker-independent splits—the usual benchmark hygiene in gesture and action recognition—are necessary but insufficient for cross-corpus evaluation. Future benchmarks in this space should stratify test sets by *corpus of origin*, not just by speaker, and report per-corpus breakdowns alongside aggregate figures.

Insight 4: label harmonisation enables pooling but the cost is real and should be reported. Collapsing five annotation schemes into three labels (Gesture / Move / NoGesture) is what makes a 32,575-clip cross-corpus dataset possible. But the collapse is lossy in a direction that matters scientifically: the distinction between iconic, deictic, metaphoric, and beat gestures—the distinctions that drove the collection of most of these corpora in the first place—disappears into the single GESTURE class. Models that perform well on this benchmark are detecting *communicative hand movement*; they are not detecting gesture in the richer sense used in any of the contributing corpora. The benchmark is a floor, not a ceiling. We encourage Phase 2 participants to report fine-grained per-category performance on corpora where such labels survive the harmonisation.

Challenge infrastructure: current state. The EnvisionBOX evaluation pipeline is available as a set of command-line tools: `check_submission.py` (format validation), `evaluate_submission.py` (balanced accuracy, per-class F1, confusion matrix), and `leaderboard.py` (ranking across submissions), built on the MaskBench framework (<https://maskbench.github.io/maskbench/>). The submission portal (CodaLab or EvalAI) will be finalised before Phase 2 opens in Q4 2026;

a web-based challenge generator UI for corpus donors to submit new kinematic datasets is on the roadmap for a subsequent round. Starter kit, evaluation script, and training data are available at the challenge repository linked from the MaskBench project site.

The lab–wild and cross-linguistic divides. ECOLANG (54% of training clips) and TEDM3D index naturalistic speech; GESRes and SaGA index controlled lab settings. Models that excel in one condition may fail in the other, and the interaction with language (gesture rate, form, and pragmatic function) differ across English, German, and Mandarin subcorpora (Hostetter and Alibali, 2008)) is entirely uncharted at cross-corpus scale. Phase 2 leaderboards will report per-corpus and per-setting breakdowns to make these gaps visible.

9 Conclusion

We introduced EnvisionBOX, the first multi-corpus shared benchmark for co-speech gesture detection, built on anonymised world landmarks extracted by the MaskBench framework. Privacy-preserving kinematic extraction enables seven independent laboratories to donate data they could not otherwise share, yielding a dataset—32,575 clips, three languages, five interaction settings, 150+ speakers—that is an order of magnitude broader in scope than any prior gesture detection benchmark. Two strong baselines set a competitive starting point, but the 6-percentage-point gap between cross-validated and held-out balanced accuracy demonstrates that cross-corpus gesture detection is a genuinely open challenge. We invite the MINT community to benchmark on EnvisionBOX and collectively build the shared infrastructure that gesture research needs.

Limitations

Ground truth labels reflect each corpus’s original annotation conventions, harmonised post-hoc into three classes. Label boundaries and gesture definitions vary across labs, introducing noise that may be larger than annotation disagreement within a single corpus. The current baseline models treat detection as a clip-level classification problem; online segmentation from continuous streams is not evaluated here. The kinematic information bottleneck—world landmarks only—limits comparability with prior video-based gesture recognition work that uses full visual appearance. Speaker counts and per-corpus demographic statistics will be reported

in the Phase 2 publication once contributor verification is complete (Issue #26).

Ethics Statement

All data was collected with informed consent under the ethical frameworks of the contributing institutions. Only anonymised kinematic data is distributed: world landmark coordinates and rendered skeleton animations. No raw video, audio, face mesh coordinates, or raw pixel data are released. ECOLANG contains adult speakers only; the data provider has confirmed that the anonymised derivatives cannot be linked back to original recordings. MULTISIMO is restricted to non-commercial use per its original licence. All EnvisionBOX data is released under CC BY-NC-SA 4.0. Any derivative work must cite the original corpus papers and credit all data contributors as co-authors.

Acknowledgements

Contributor names and affiliations for all seven corpora are being verified with each data owner before publication (Issue #26 in the `envision-gesture-challenge` repository). This section will credit all corpus providers and acknowledge funding sources on camera-ready submission. **[TODO: Add funding acknowledgements: NWO, BMBF, etc. as applicable.]**

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